

FinTrust Digital Bank

Week 1 Report — Understand & Plan

FinTrust Financial Intelligence & Digital Banking Support Solution

AnalystLab Africa — Experience Lab Internship Programme

Field	Detail
Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Track	Data Analytics (shared evidence base for all five tracks)
Week / Version	Week 1 — Understand & Plan · v1.0
Submission deadline	End of Week 1 (Sunday, 11:59 PM WAT)
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1. Executive Summary

Analysis of the approved datasets established the core facts: 1,500 customers and 12,000 transactions covering 1 January to 31 March 2026, with a clean join on Customer_ID (0 orphan transactions, 8.0 transactions per customer). Total transaction value is NGN 560,477,355, the transaction success rate is 90.47%, and 19.6% of transactions (2,352) are flagged for risk review, carrying NGN 161,783,631 — 28.9% of all value.

Four statistically significant risk drivers were confirmed: international transactions (36.88% vs 18.88% flagged), transaction amount (NGN 68,786 vs 41,324), Failed status (26.98%) and night-time execution (00:00–05:00, ~28%). A baseline logistic-regression classifier achieved ROC-AUC 0.673 — it ranks transactions correctly but, at the default threshold, collapses onto the majority class (recall 0.039). This is reported honestly and becomes the Week 2 improvement target.

2. Week 1 Objectives & Success Criteria

2.1 Objectives

- Understand the FinTrust business problem and how data can address it.
- Review and validate every approved resource (customer, transaction, data dictionary).
- Define track-specific outputs: profiling report, KPI table, analytical answers, dashboard.
- Identify the questions, KPIs, features and risks relevant to the track.
- Develop and document a credible approach and plan for Weeks 2–4.

2.2 Success criteria

- Outputs are well organised, clearly documented, professional and evidence-backed.
- Every KPI and claim is reproducible from the supplied data.
- The risk model must beat the 0.804 do-nothing baseline in Week 2.

- Outputs are relevant to the FinTrust business problem and easy for a reviewer to evaluate.

3. Business Understanding

What FinTrust is trying to solve. FinTrust Digital Bank is a fictional digital bank that cannot yet systematically convert its customer and transaction data into decisions. It wants to (1) understand customer and transaction behaviour, (2) support risk-related decision-making, (3) improve operational processes, and (4) strengthen digital customer support.

Why it matters. The bank holds NGN 560,477,355 of transaction value. Nearly one fifth of transactions sit in a manual risk-review queue holding NGN 161,783,631, while 7.97% of transactions fail or reverse and 8.86% of customers are dormant or restricted. Each of these represents measurable cost, friction or exposure.

How this track contributes. Data Analytics converts the raw behaviour into visible KPIs, cross-tabulated insight and a decision-ready dashboard, providing the evidence base the Data Science, ML Engineering, GenAI and Project Management tracks build upon.

Output type. Decision artefacts: a validated data profile, a KPI table, an analytical insight pack, a dashboard wireframe and an initial analysis plan.

4. Data & Resource Understanding

4.1 Resource inventory and limitations

Resource	Verified contents	How used	Limitation identified
FinTrust_Customer_Data.xlsx	1,500 records x 12 fields; 0 missing; 0 duplicates; key Customer_ID (1,500 unique)	Customer segmentation, lifecycle analysis, dimension table for joins	Customer_Name repeats (180 unique names for 1,500 customers) — not an identifier
FinTrust_Transaction_Data.xlsx	12,000 records x 11 fields; 192 missing cells; 0 duplicate rows; key Transaction_ID (12,000 unique)	Behaviour, value and the risk target	96 rows missing Device_Type; 96 missing Location
FinTrust_Data_Dictionary.xlsx	12 field definitions with data type, definition, business use and modelling-use flag	Schema contract; validates which fields are safe to model	Explicitly marks Customer_Name as Modelling_Use = No (identification only)
FinTrust Knowledge Base / Brief / Roadmap	Policy and scope context	GenAI grounding; PM scope control	

4.2 Data structure (verified)

Attribute	Customer file	Transaction file
Records	1,500	12,000
Columns	12	11
Primary key	Customer_ID (1,500 unique)	Transaction_ID (12,000 unique)
Duplicate rows	0	0
Missing cells	0	192 (Device_Type 96; Location 96)
Date coverage	not applicable	2026-01-01 to 2026-03-31 (90 days)
Numeric fields	Age, Tenure_Months, Digital_Engagement_Score	Amount_NGN

Categorical fields	Gender(3), City(8), Customer_Segment(4), Account_Type(3), Monthly_Income_Band(5), Preferred_Channel(3), Account_Status(3)	Transaction_Type(6), Channel(5), Device_Type(5), Location(8), International_Transaction(2), Transaction_Status(4), Risk_Review_Flag(2)
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4.3 Relationship and join integrity

The customer file is the dimension table and the transaction file is the fact table; they join on Customer_ID. Verification confirmed 1,500 distinct customers in both files, 0 orphan transactions, 0 customers without transactions, and a mean of 8.00 transactions per customer (minimum 1, maximum 20).

4.4 Data-quality log

Check	Result	Issue	Action
Missing cells (transaction)	192	96 Device_Type + 96 Location	Impute 'Unknown' or exclude the 96 affected rows
Customer_Name uniqueness	180 distinct names for 1,500 customers	Repeats — not an identifier	Join and identify on Customer_ID only
Transaction.Location vs Customer.City	12.7% agreement	Same 8 labels but different meaning	Do NOT join; treat as separate fields
Amount distribution	skewness 3.2	Heavy right tail	Apply a log transform before linear modelling
Duplicate rows	0	None	No action

5. Methodology

- Loaded all three approved workbooks with pandas; confirmed shape, dtypes, missingness, duplicates and primary keys.
- Validated the Customer_ID relationship between the fact and dimension tables and quantified per-customer activity.
- Computed the bank-wide KPI set (value, success, failure, risk, engagement).
- Produced cross-tabulations of the risk-review rate across ten dimensions.
- Tested four hypotheses using chi-square tests of independence and Welch t-tests.
- Built a reproducible baseline logistic-regression classifier (stratified 75/25 split, random_state=42).
- Rendered the analysis figures and the dashboard/architecture wireframes.
- Packaged everything as a notebook, scripts, a CSV/JSON evidence pack and this report.

6. Findings — Key Performance Indicators

KPI	Value	Notes
Total Transaction Value	NGN 560,477,355	SUM of Amount_NGN
Total Transactions	12,000	12,000
Total Customers	1,500	1,500
Average Transaction Amount	NGN 46,706.45	mean
Median Transaction Amount	NGN 10,305.94	median
95th Percentile Amount	NGN 227,285.52	p95
Transaction Success Rate	90.47%	Successful / total
Failed + Reversed Rate	7.97%	operational friction

Risk-Review Flag Rate	19.6%	2,352 transactions
Value Under Risk Review	NGN 161,783,631	28.9% of total value
International Transaction Share	4.0%	480 transactions
Active Account Rate	91.13%	1,367 of 1,500
Average Digital Engagement Score	68.0/100	0–100 scale
Average Tenure	49.9 months	customer lifecycle

7. Findings — Cross-Tabulations

7.1 Risk-review rate by Channel

Category	Transactions	Risk flagged	Risk rate %	Average amount (NGN)
Web	1,869	399	21.35%	47,793
ATM	1,747	370	21.18%	48,049
Mobile App	5,102	990	19.4%	47,061
POS	2,393	439	18.35%	43,605
USSD	889	154	17.32%	48,098

7.2 Risk-review rate by Transaction Type

Category	Transactions	Risk flagged	Risk rate %	Average amount (NGN)
Transfer	3,549	1,011	28.49%	67,038
Cash Withdrawal	1,430	362	25.31%	47,393
Deposit	1,328	215	16.19%	98,633
Airtime/Data	1,185	180	15.19%	8,250
Card Purchase	3,033	395	13.02%	28,310
Bill Payment	1,475	189	12.81%	19,094

7.3 Risk-review rate by Transaction Status

Category	Transactions	Risk flagged	Risk rate %	Average amount (NGN)
Failed	630	170	26.98%	39,406
Successful	10,856	2,089	19.24%	47,053
Reversed	326	60	18.4%	47,933
Pending	188	33	17.55%	49,032

7.5 Risk-review rate by Domestic vs International

Category	Transactions	Risk flagged	Risk rate %	Average amount (NGN)
Yes	480	177	36.88%	41,658
No	11,520	2,175	18.88%	46,917

7.6 Risk-review rate by Customer Segment

Category	Transactions	Risk flagged	Risk rate %	Average amount (NGN)
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Premium	2,361	480	20.33%	45,638
Student	2,289	452	19.75%	46,953
SME	1,706	331	19.4%	49,116
Everyday	5,644	1,089	19.29%	46,325

8. Findings — Hypothesis Tests

Four hypotheses were stated in the Week 1 plan and tested formally. Chi-square tests of independence were used for categorical dimensions; a Welch t-test for the continuous amount variable.

Hypothesis	Test	p-value	Result
H1 International transactions are more likely flagged	chi-square = 93.55	3.96e-22	Supported (36.88% vs 18.88%)
H2 Higher-value transactions are more likely flagged	Welch t = 11.30	5.33e-29	Supported (NGN 68,786 vs 41,324)
H3 Failed transactions are more likely flagged	chi-square = 23.47	3.22e-05	Supported (26.98%)
H4 Night-time transactions (00:00–05:00) are more likely flagged	chi-square = 194.34	3.59e-44	Strongly supported (~28% vs ~17%)

Dimensions that were NOT significant: Device_Type (p=0.156), Customer_Segment (p=0.752), Account_Status (p=0.462), Monthly_Income_Band (p=0.467), Preferred_Channel (p=0.835) and City (p=0.834). These should not be treated as risk drivers on their own evidence.

9. Baseline Predictive Model (Data Science Track)

Metric	Value	Interpretation
Target	Risk_Review_Flag	Yes / No; 19.6% positive (4:1 imbalance)
Algorithm	Logistic Regression	deliberately simple baseline
Validation	Stratified 75/25 split	random_state=42; test = 3,000 rows
ROC-AUC	0.673	ranks correctly, better than chance
Do-nothing baseline accuracy	0.804	a model predicting all-negative
Model accuracy @0.50	0.801	no better than do-nothing at this threshold
Precision (positive class)	0.418	of predicted flags, 42% correct
Recall (positive class)	0.039	only 4% of true flags caught
Confusion matrix [TN, FP, FN, TP]	2380 / 32 / 565 / 23	threshold too high for imbalanced target

Top standardised coefficients: International_Transaction +0.93, IsNight +0.77, Transaction_Type_Transfer +0.67, Transaction_Type_Cash Withdrawal +0.66, Amount_NGN +0.25.

Honest reading. The model ranks transactions correctly (AUC 0.673 > 0.50) but at the default 0.50 threshold it collapses onto the majority class. The reported accuracy of 0.801 is therefore not evidence of skill — it equals the do-nothing baseline. Week 2 fixes: class_weight='balanced', threshold tuning, and gradient-boosted trees, evaluated with PR-AUC and recall@k rather than accuracy.

10. Dashboard Design (Data Analytics Track)

Region	Content
Filter bar	Date range, City, Customer Segment, Account Type, Channel, Transaction Type, Status, Risk flag
KPI cards	Total transactions, total value, success rate, risk-review rate, failed+reversed rate, value under risk review
Chart 1	Transaction volume and value trend by month
Chart 2	Risk-review rate by channel / device (sorted descending)
Chart 3	Risk-review rate by transaction type
Chart 4	Customer base: segment, city and engagement score
Insights panel	Top risk channel, peak failure month, high-value customers, international exposure, dormant/restricted accounts

11. ML Engineering Architecture

Workflow: Data → Validation → Pre-processing → Feature Preparation → Model → Prediction → Output (API) → Testing.

- Technical requirements: schema-validated inputs; pre-processing serialised with the model to guarantee train/serve parity; pinned requirements.txt; Git version control; FastAPI POST /predict with Pydantic validation; batch-scoring fallback.
- Repository structure: /data (raw, interim, processed), /notebooks (exploration only), /src (data_validation.py, preprocess.py, features.py, train.py, predict.py), /models (versioned artefacts), /tests (pytest), /configs, /api, Makefile, requirements.txt.
- Technical risks: missing Device_Type/Location; train-serve mismatch; dependency drift; invalid values; API versioning failure; non-reproducibility; class imbalance masked by accuracy.
- Testing: unit tests per function, schema tests on load, data-drift checks, a model-metric regression gate, and API contract tests.

12. Generative AI Scope & Architecture

Supported topics: account access, failed transactions, reversals, transfer troubleshooting and security guidance as documented in the approved FinTrust knowledge base. Unsupported and must escalate: bespoke fees, limits, timelines and account-specific balances.

Use cases: resolve account-access issues; explain a failed transaction; explain reversals; troubleshoot transfers; general security guidance.

Exclusions (must never do): invent fees, limits or timelines; give financial advice; request PIN/OTP/credentials; execute transactions; disclose account-specific data without authentication.

Architecture: User Question → Prompt & Guardrail → Retrieval (knowledge-base search) → LLM (answer only from retrieved context) → Grounded Response with citation → Evaluation/Hallucination Guardrail. If retrieval finds no support, the assistant must refuse and escalate to a human agent. Every question, answer and retrieved chunk is logged for audit, and account numbers are masked.

13. Project Management — Risks & Delivery

Risk	Probability	Impact	Mitigation
Data-quality gaps (192 cells)	Medium	Medium	Validation rules at ingestion
Class imbalance hides failure	High	High	Balanced metrics and threshold tuning
Train-serve feature mismatch	Medium	High	Shared serialised pre-processor
Hallucinated policy answers	Medium	High	Knowledge-base-only grounding plus guardrail
Scope creep	Medium	Medium	Charter scope lock
Late cross-track integration	Medium	Medium	Cross-track sync begins Week 2
Non-reproducible pipelines	Medium	High	Seeded, containerised, pinned dependencies
Single-person dependency	Low	High	Documented handover
Submission links inaccessible	Low	High	Test every link before submitting

14. Plan for Weeks 2–4

Week	Deliverables
Week 2	Data Analytics: build the dashboard and the analytical SQL queries. Data Science: engineered features and a tuned, balanced model. ML Engineering: training-pipeline skeleton. GenAI: knowledge-base chunking and the first prompt set. Project Management: risk-register update.
Week 3	Data Science: evaluated model served behind the API. ML Engineering: FastAPI /predict endpoint plus tests. Data Analytics: deep-dive on risk and channel behaviour. GenAI: retrieval evaluation run against the 10-question test set.
Week 4	Integration across tracks, full documentation, executive summary, final presentation and submission.

15. Assumptions & Limitations

- All FinTrust customers, transactions and financial information are synthetic and created for educational purposes; no real personal data is involved.
- Risk_Review_Flag is a synthetic label of unknown provenance and may not reflect a real-world risk definition.
- Transaction.Location and Customer.City are distinct concepts despite sharing category labels.
- The 96 rows missing Device_Type and Location were retained for profiling; they must be imputed or excluded before modelling.
- The baseline model's threshold behaviour reflects class imbalance, not a modelling failure; tuning is deferred to Week 2.

Appendix — Submitted Artefacts

Artefact	Description
FinTrust_Week1_Report .pdf	This report
FinTrust_Week1_Dashboard.xlsx	Excel dashboard with KPI table, cross-tabs, hypothesis tests, model baseline, data-quality log and native charts
FinTrust_Week1_Analysis.ipynb	Reproducible Jupyter/Colab notebook

analysis.py / profile.py	Analysis and profiling scripts
FinTrust_KPI_Table.csv / results.json	Evidence pack of computed values
fig1–fig4 PNG	EDA charts, dashboard wireframe, ML architecture, GenAI architecture
GitHub repository	Source code and documentation